Visual Analytics Homework

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1 Visualization of Bechelor Exams

In this section, we classify the visualizations presented below, considering both individual (Federico) and group (4 students) cases.

1.1 Individual Performance: Federico

1.1.1 Bar Chart of Federico's Grades

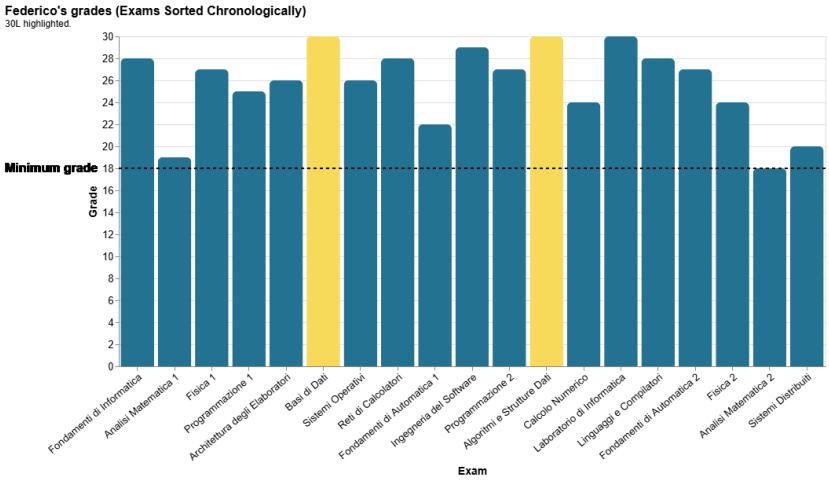


Figure 1: Bar Chart of Federico's Grades.

- Visual Task: Comparison – allows comparison of grades across exams.

- Type of Data:
  - Nominal: exam names (X-axis).
  - Nominal (derived): honors (30L) highlighted in yellow.
  - Quantitative: numeric grades (Y-axis).
- Relationships:
  - Each exam corresponds to one grade.
  - Honors (30L) are visually distinguished from normal grades.
- Visual Variables / Variations:
  - Position (x-axis): exams name.
  - Position (y-axis): quantitative, grade value.
  - Color: nominal, highlights 30L.

1.1.2 Temporal Trend of Federico's Grades

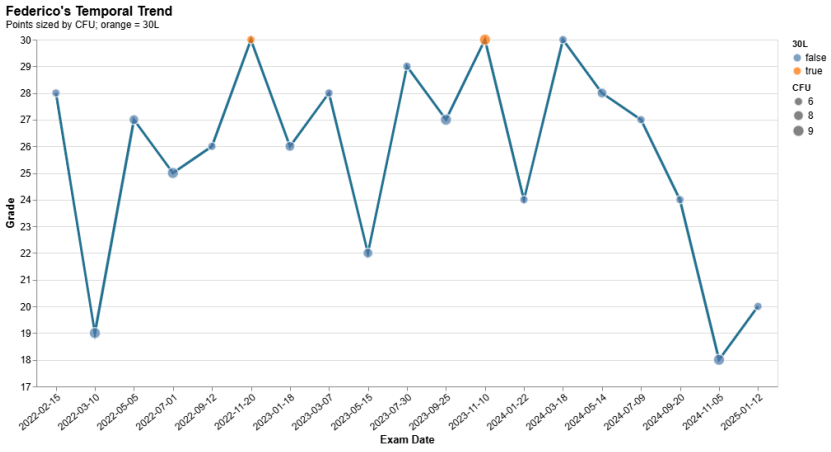


Figure 2: Line Chart for Federico's Grades.

- Visual Task: Trend / Change over Time – shows how grades evolve temporally.

  - Temporal: exam dates (X-axis).
  - Quantitative: numeric grades (Y-axis), CFU (point size).
  - Nominal (derived): honors (30L) highlighted with yellow points.

Relationships:

- Exam date → grade (temporal progression).
- CFU → point size.
- Honors → point colored with yellow.

Visual Variables / Variations:

- Position (x-axis): temporal (exam date).
- Position (y-axis): quantitative (grade).
- Color: nominal (30L vs normal).
- Size: quantitative (CFU).
- Line / Point Overlay: shows individual temporal trends.

1.1.3 Bubble Chart for Federico's Grades

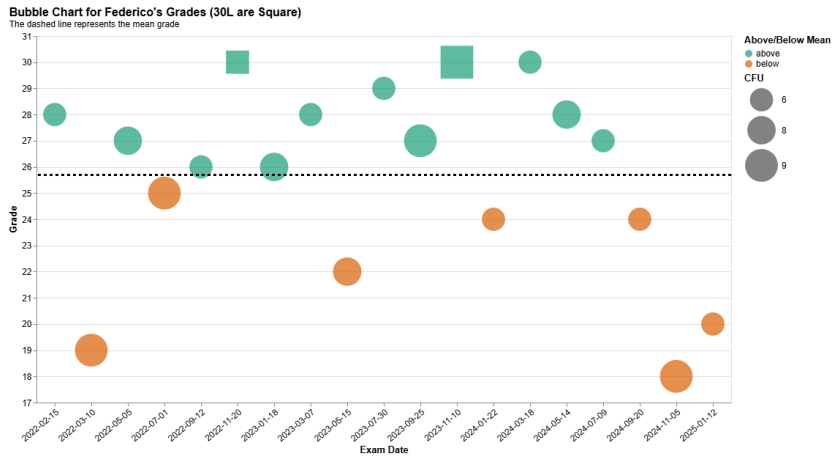


Figure 3: Bubble Chart for Federico's Grades.

Visual Task: Comparison – allows for comparison of grades across exams, including size (CFU) and honors (30L).

Type of Data:

- Temporal: exam dates (X-axis).
- Quantitative: numeric grades (Y-axis).
- Quantitative: CFU of each exam (bubble size).
- Nominal (derived): honors (30L) highlighted as square shape.

Relationships:

- Each exam corresponds to one numeric grade.
- Honors (30L) are visually distinguished with a square.
- Bubble size encodes the CFU for each exam.
- Above/below mean grade is represented with color coding (green = above mean, orange = below mean), while 30L are highlighted as square.

Visual Variables / Variations:

- Position (X-axis): temporal order of exams.
- Position (Y-axis): quantitative, grade value.
- Size: quantitative , representing CFU.
- Shape: nominal, distinguishes 30L (square) vs normal grades (circle).
- Color: nominal, distinguishes above/below mean and 30L.

1.2 Group Performance: 4 Students

1.2.1 Box Plot of Group Grades

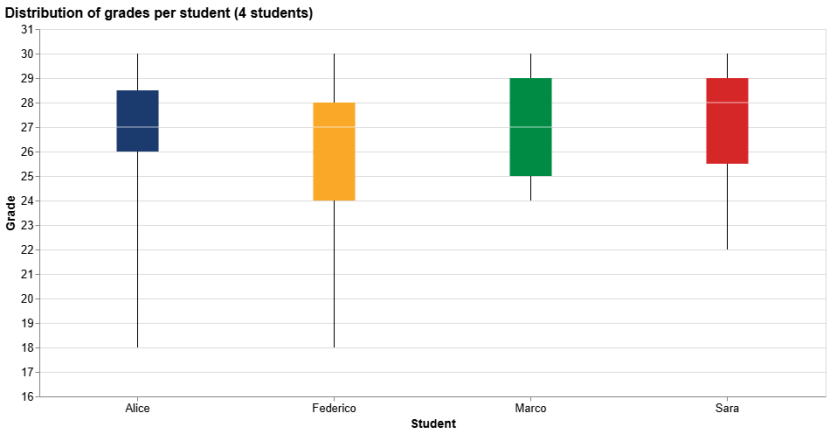


Figure 4: Box Plot for Group.

- Visual Task: Comparison and Distribution – compares grades across students and shows spread.
- Type of Data:
  - Nominal: student names.
  - Quantitative: grades.

- Relationships:
  - Each student $\rightarrow$ distribution of grades.
- Visual Variables / Variations:
  - Position (x-axis): nominal, student.
  - Position (y-axis): quantitative, grade.
  - Color: nominal, distinct per student.
  - Size / Shape: boxplot elements encode distribution statistics.

1.2.2 Temporal Trend of Federico vs Group Average Grades

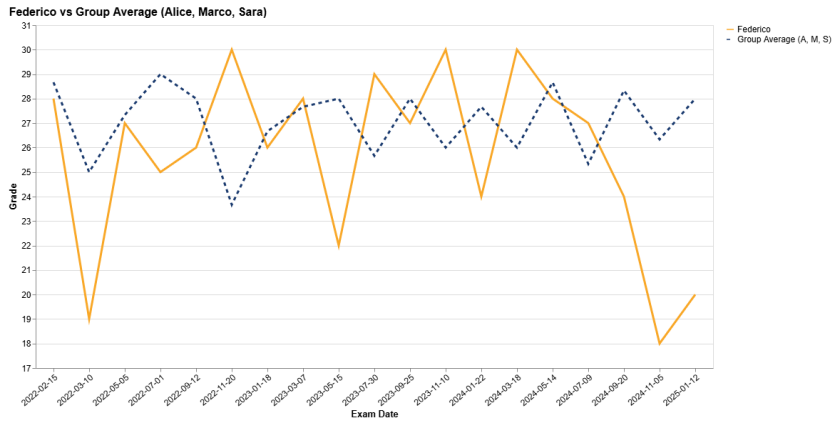


Figure 5: Line Chart for Federico vs Group Average Grades.

- Visual Task: Trend / Change over Time for individual vs group comparison.
- Type of Data:
  - Temporal: exam dates (X-axis).
  - Quantitative: grades (Y-axis).
  - Nominal: line identity (yellow for Federico, blue for group average of Alice, Marco, Sara).
- Relationships:
  - Group average (Alice, Marco, Sara) → aggregated mean per exam date.
  - Federico's trend → directly compared to the group average.
- Visual Variables / Variations:
  - Position (x-axis): temporal, exam date.

- Position (y-axis): quantitative, grade.
- Color: nominal, distinguishes Federico from the group average.
- Dashed Line: differentiates Federico's line (solid) from the group average (dashed).

1.2.3 Temporal Trend for every Student

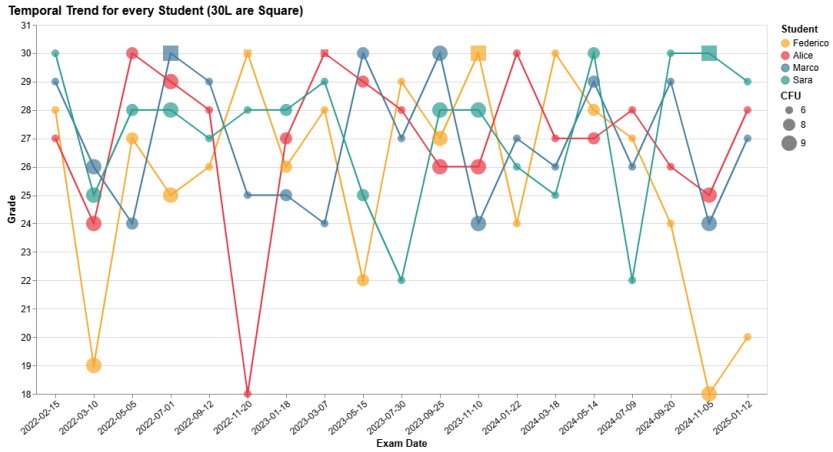


Figure 6: Line Chart for Federico vs Group.

- Visual Task: Temporal Trend – allows comparison of grades across exams and multiple students over time, including CFU and honors (30L).
- Type of Data:
  - Temporal: exam dates (X-axis).
  - Nominal: student names (encoded as color).
  - Quantitative: numeric grades (Y-axis).
  - Quantitative (represented as bubble size): CFU of each exam.
  - Nominal (derived): honors (30L) highlighted as square shape.

- **Relationships:**
  - Each exam corresponds to one grade.
  - Honors (30L) are visually distinguished with a square shape.
  - Bubble size encodes the CFU for each exam.
  - Color encodes student identity.
  - Position along X-axis follows chronological exam date.
- **Visual Variables / Variations:**
  - Position (X-axis): exam dates (temporal order).
  - Position (Y-axis): quantitative, grade value.
  - Size: quantitative, representing CFU.
  - Shape: nominal, distinguishes 30L (square) vs normal grades (circle).
  - Color: nominal, distinguishes students.

2 Expected and Unexpected Insight from Gapminder

2.1 Expected Insight

2.1.1 Wealth and Happiness

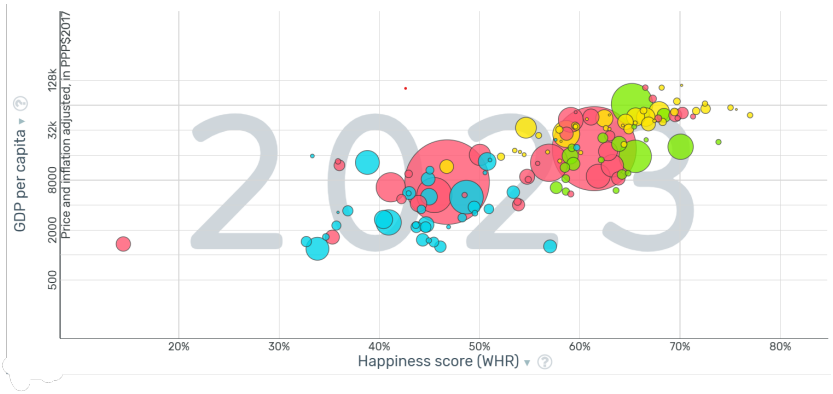


Figure 7: Wealth and Happiness Chart.

As expected, this chart shows a positive and usual correlation: countries with higher GDP per capita (higher on the Y-axis) also report higher Happiness Scores (X-axis).

$$\text{GDP per capita (Wealth)} \uparrow \implies \text{Happiness Score} \uparrow$$

This result is expected because it is intuitive to think that national wealth provides citizens with a better quality of life, through superior access to healthcare, education, safety, and stability. These factors form the basis for greater well-being and, therefore, a higher happiness score.

2.1.2 Happiness and Work-Life

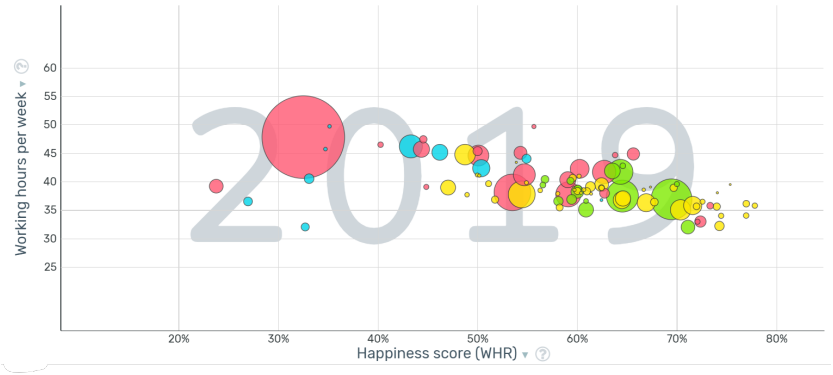


Figure 8: Happiness and Work-Life Chart.

This chart confirms a intuition: the happiest countries are those where people work less. A clear trend is observed: as the Happiness Score (X-axis) increases, Weekly Working Hours (Y-axis) decrease.

$$\text{Happiness Score} \uparrow \implies \text{Weekly Working Hours} \downarrow$$

This result is expected because work-life balance is a crucial factor for well-being. Having more free time for family, hobbies, and rest, thanks to shorter working hours, is directly linked to greater personal satisfaction and, consequently, a higher happiness score.

2.1.3 Poverty as an Obstacle to Education

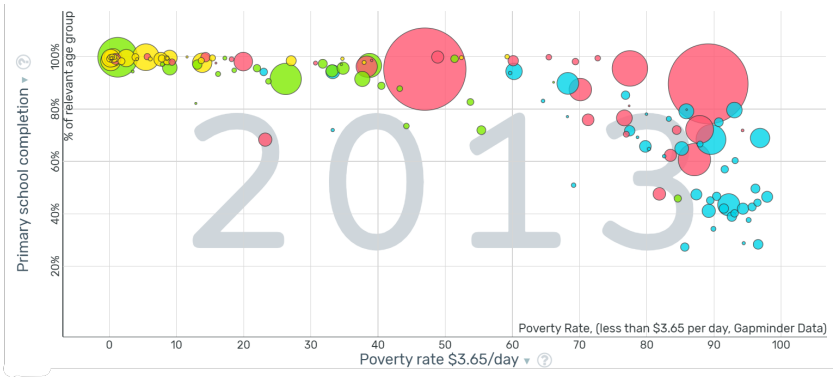


Figure 9: Poverty and Education Chart.

This chart shows a strong and usual negative correlation: as the Poverty Rate (X-axis) increases, the Primary School Completion Rate (Y-axis) drops sharply.

$$\text{Poverty Rate} \uparrow \implies \text{Primary School Completion} \downarrow$$

This is expected because countries with low poverty (on the left) have almost universally achieved full school completion (close to 100%). In countries with high poverty (on the right), the need to work, lack of infrastructure, and limited family resources prevent many children from completing primary education.

2.2 Unexpected Insight

2.2.1 Fertility and Wealth

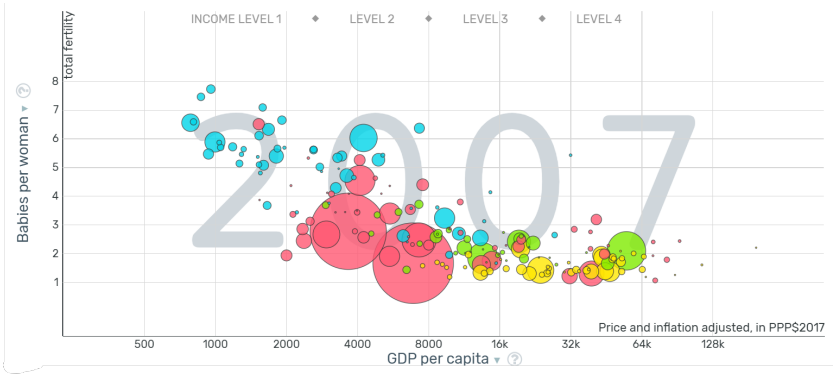


Figure 10: Fertility and Wealth Chart.

The chart shows a counterintuitive correlation.

$$\text{Wealth (GDP per capita)} \uparrow \implies \text{Number of children (Fertility)} \downarrow$$

The analysis highlights a paradox: one might expect that higher economic availability and security would encourage or enable larger families. The data, however, show the exact opposite.

Countries with the highest incomes almost universally have lower fertility rates, often below the replacement level (≈ 2.1), while the highest rates are found in countries with lower incomes.

2.2.2 Work and Wealth

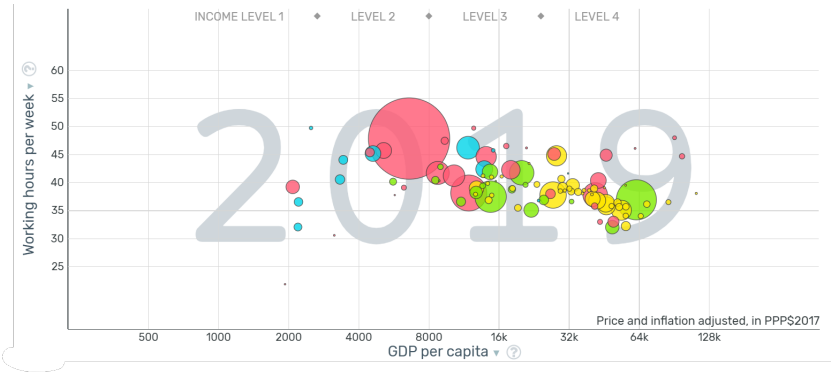


Figure 11: Work and Wealth Chart.

This chart shows a striking correlation: wealthier countries (with higher GDP per capita) are those where citizens work, on average, fewer hours per week (lower on the Y-axis).

$$\text{GDP per capita} \uparrow \implies \text{Weekly Working Hours} \downarrow$$

This pattern is unexpected because it challenges common assumptions. Usually, national wealth is associated with working a lot. This chart, however, could suggests that wealth does not come from the quantity of hours worked, but from high productivity, which allows rich countries to generate more GDP in less time.

2.2.3 Poverty and Number of Suicides

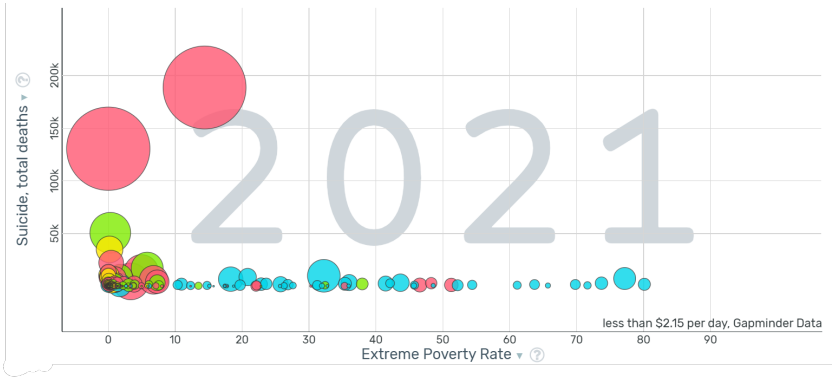


Figure 12: Poverty and Number of Suicides Chart.

Contrary to what one might expect, this chart reveals an inverse relationship between poverty and the total number of suicides. Wealthier countries (with lower poverty rates on the X-axis) tend to have higher total numbers of suicides (higher on the Y-axis), while poorer countries show no consistent increase in suicides despite higher poverty levels.

$$\text{Poverty Rate} \downarrow \implies \text{Suicide Numbers} \uparrow$$

This finding is striking because it contradicts the common belief that extreme poverty leads directly to higher suicide rates. Instead, the data indicate that many wealthy nations (with poverty rates close to zero) display higher total suicide numbers than almost all of the poorest countries. The determining factor in this chart appears to be **population size**: countries with larger populations such as China and India naturally report higher absolute numbers of suicides, regardless of poverty levels.